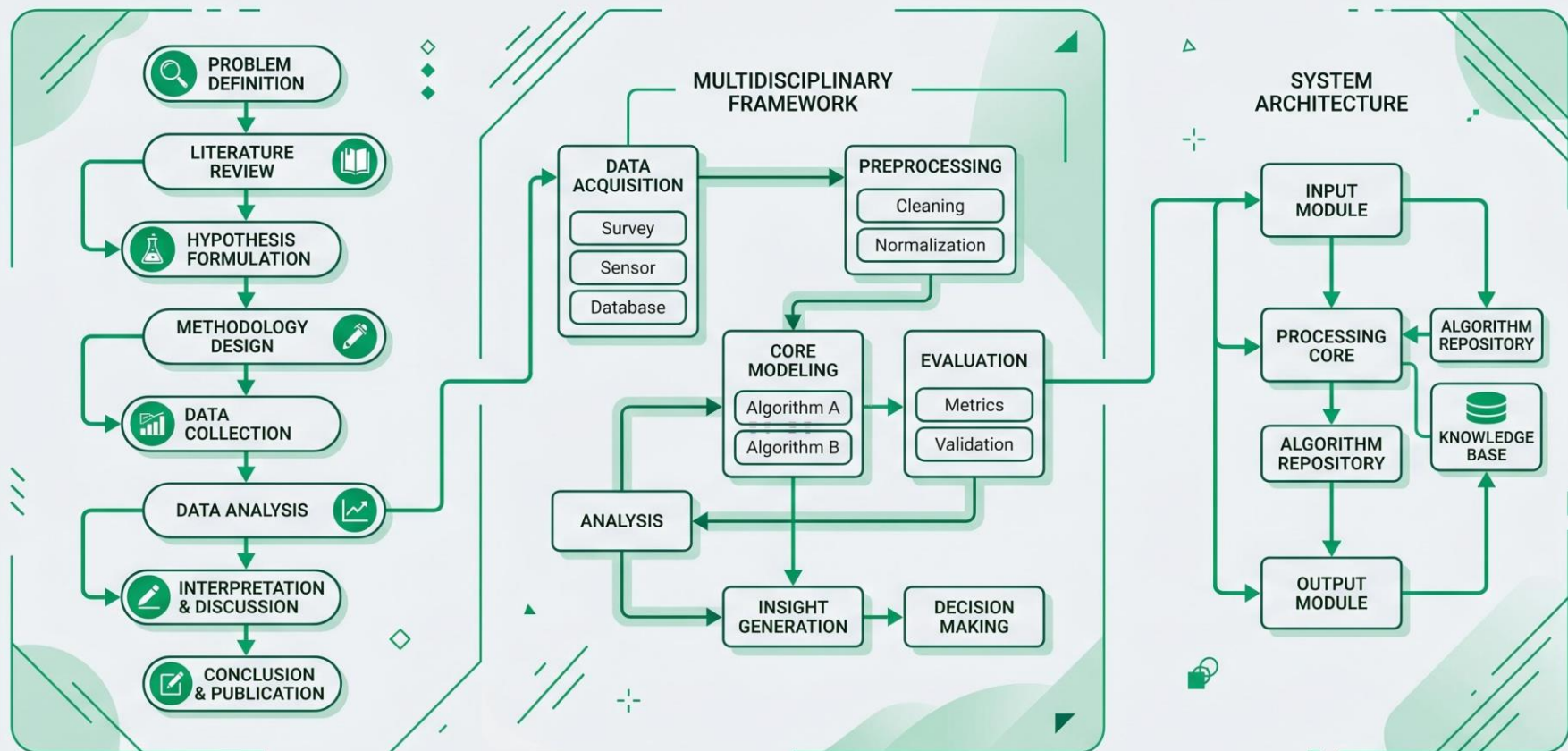


SMBFLOW — RESEARCH LITERATURE REVIEW & PROPOSED PROJECT DIRECTION

IEEE Literature Review (2025–2026) and Proposed System Development

Prepared for project research and faculty review



1. Research Objective

The project aims to develop an intelligent workflow automation platform for small and medium-sized businesses that combines predictive machine learning, explainable AI, large language model reasoning, workflow orchestration, human oversight and measurable business outcomes.

The proposed approach focuses on moving from simple task automation toward outcome-oriented automation: the system first identifies a business situation or risk, predicts its likely outcome, explains the prediction, recommends an appropriate action, executes the action when permitted, and records the resulting outcome for evaluation and future improvement.

2. Literature Review — IEEE Papers (2025–2026)

The following papers were selected to cover the technical foundations of the proposed system: predictive business-process monitoring, Agentic AI, multi-agent workflows, explainable ML, human-in-the-loop decision making, enterprise data access, API/tool execution, provenance and operational optimization.

Sr. No.	Title of Research Paper	Authors	Name of Journal / Conference	Date of Publication	Brief Summary of Work (2/3 lines)	Remarks (Key Findings / Important Technology / Tools / Software Used)	IEEE Xplore URL
1	Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey	Deepak Bhaskar Acharya; Karthikeyan Kuppan; B. Divya	IEEE Access	22 January 2025	Surveys Agentic AI systems designed to pursue complex goals with limited human intervention. It covers autonomy, decision-making, adaptation, architectures and deployment challenges.	Agentic AI; autonomous agents; reinforcement learning; adaptive control. Key issues include reliability, explainability, resource constraints and governance.	https://ieeexplore.ieee.org/document/10849561/
2	Agentic AI: A Survey of Autonomous Agents, Architectures, and Emerging Applications	Donghun Lee; Hyosu Kim	2025 IEEE ICTBIG	12–13 December 2025	Reviews autonomous-agent architectures from classical agents to LLM-powered and hybrid systems, including perception, planning, action and collaboration.	BDI agents; reinforcement-learning agents; LLM agents; multi-agent systems. Highlights reliability, explainability, coordination and scalability.	https://ieeexplore.ieee.org/document/11323264/
3	Alvaluate: A Multi-Agent Framework for Automated Answer Script Evaluation Using Large Language Models and Semantic Vector Indexing	P. Suryakumar; A. Malini; S. K. Subasini; G. Priyanka; et al.	IEEE Access	2025	Uses specialized agents, LLMs and semantic-vector techniques in a modular workflow for automated evaluation.	LLMs; semantic vector indexing; Azure Document Intelligence; Azure AI Foundry; MongoDB; FAISS; embeddings. Supports modular agent orchestration.	https://ieeexplore.ieee.org/document/11155081/
4	MAD-CTI: Cyber Threat Intelligence Analysis of the Dark Web Using a Multi-Agent Framework	Sayuj Shah; Vijay K. Madiseti	IEEE Access	3 March 2025	Uses specialized LLM agents to collect, analyse and classify cyber-threat intelligence. The work decomposes a complex analysis task into multiple cooperating agents.	LLM agents; scraping; translation; classification; multi-agent collaboration. Supports specialized-agent workflow design.	https://ieeexplore.ieee.org/document/10908603/
5	Towards Detecting LLMs Hallucination via Markov Chain-based Multi-agent Debate Framework	Xiaoxi Sun; Jinpeng Li; Yan Zhong; Dongyan Zhao; Rui Yan	ICASSP 2025	6–11 April 2025	Proposes a multi-agent debate approach for identifying hallucinated LLM claims using verification and evidence-oriented reasoning.	Multi-agent debate; Markov chains; evidence retrieval; LLM verification. Supports verification and confidence-aware reasoning.	https://ieeexplore.ieee.org/document/10889448/
6	AnaFlow: Agentic LLM-based Workflow for Reasoning-Driven Explainable and Sample-Efficient Analog Circuit Sizing	Mohsen Ahmadzadeh; Kaichang Chen; Georges Gielen	2025 IEEE/ACM ICCAD	26–30 October 2025	Uses specialized LLM agents that collaborate, iteratively refine solutions and learn from optimization history. The workflow combines reasoning,	Multi-agent LLM workflow; adaptive simulation; optimization history; explainable reasoning. Demonstrates an iterative agentic workflow.	https://ieeexplore.ieee.org/document/11240818/

					execution and explanation.		
7	Generative Explainability for Next-Generation Networks: LLM-Augmented XAI with Mutual Feature Interactions	Kiarash Rezaei; Omran Ayoub; Sebastian Troia; Francesco Lelli; Paolo Monti; Carlos Natalino	2025 IEEE WiMob	20–22 October 2025	Combines SHAP feature influence and feature interactions with an LLM to generate human-understandable explanations of ML predictions.	SHAP; XAI; mutual feature interactions; LLMs. Reported improved explanation usefulness and correctness. Strong support for ML → XAI → LLM explanation.	https://ieeexplore.ieee.org/document/11257542/
8	A Human-in-Multi-Agent-Loop Intent Refinement Method for Task Offloading in Mobile Edge Computing	Shao Yuan; Weichen Ni; Boxiao Han; Zuojun Dai; et al.	2025 IEEE ECIS	23–25 May 2025	Integrates human judgement into a multi-agent LLM workflow and repeatedly uses feedback to refine AI outputs and decisions.	LLM workflows; multi-agent collaboration; human feedback. Supports approval, escalation and feedback mechanisms.	https://ieeexplore.ieee.org/document/11086847/
9	PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions in Agentic Workflows	Renan Souza; Amal Gueroudji; Stephen DeWitt; Daniel Rosendo; Tirthankar Ghosal; Robert Ross; Prasanna Balaprakash; Rafael Ferreira da Silva	2025 IEEE eScience	15–18 September 2025	Tracks prompts, responses, decisions and workflow context, linking agent interactions with downstream outcomes for traceability and reliability.	W3C PROV; Model Context Protocol (MCP); data observability; provenance tracking. Strong support for audit and evidence.	https://ieeexplore.ieee.org/document/11181558/
10	PDD-AGENT: Multimodal Large Language Model-Driven AI Agent for Enhanced Plant Disease Diagnosis	Lufu Qin; Xingcai Wu; Xinyu Dong; Huan Wang; Tingwei Yang; Qi Wang	2025 IEEE ICIP	14–17 September 2025	Uses separate preprocessing, decision, action and result-aggregation modules, with agents orchestrating specialized tools.	Multimodal LLM; AI agents; tool orchestration; action modules. Supports perception → reasoning → tool/action architecture.	https://ieeexplore.ieee.org/document/11084359/
11	RUBRA: An Agentic AI System for Automatic Short Answer Grading Using LLMs and RAG	Authors as listed on IEEE Xplore	2025 IEEE conference publication	2025	Combines LLMs, RAG, domain context and rubrics to produce structured scores and feedback rather than relying on unrestricted generation.	Agentic AI; LLM; RAG; semantic similarity. Supports structured context, scoring and explainable decisions.	https://ieeexplore.ieee.org/document/11189329/
12	Intent-Based Kubernetes Configuration via LLMs: Current Trends and Open Challenges	Alessio Sacco; Cristian Zilli; G. Marchetto	2025 IEEE LCN	13–16 October 2025	Studies translation of human intent into structured configuration using LLMs and identifies barriers to safe autonomous execution.	LLMs; intent-to-configuration generation; fine-tuning. Supports human intent → structured action with safeguards.	https://ieeexplore.ieee.org/document/11146359/
13	The Rise of Agentic AI in Finance: Opportunities, Risks, and Human-Centric Integration	Authors as listed on IEEE Xplore	IEEE IT Professional	14 August 2025	Examines Agentic AI for financial-service automation while discussing data quality, security, compliance and human oversight.	Agentic AI; automation; governance; human oversight. Useful for business-oriented responsible automation.	https://ieeexplore.ieee.org/document/11125703/
14	Intelligent Cybersecurity Defense Strategies Based on Large Language Model Agents	Deyang Ye; Zihao Wang; Yuyan Song; Jingtian Liang	2025 IEEE ASENS	21–23 March 2025	Develops an LLM-agent framework with decision, execution and evaluation components and compares several modern LLMs.	LLM agents; GPT-4.5; Claude 3.7; LLaMA 3.3; Gemini 2.0. Supports decision → execution → evaluation workflows.	https://ieeexplore.ieee.org/document/11011130/
15	Supply Chain with Sixth Sense Agentic AI: A Proactive Intelligence Framework for Autonomous Planning	Venkatesh Prabu Parthasarathy; Madhusudan Sharma Vadigicherla	2026 IEEE ICAIC	18–20 February 2026	Presents Perception, Cognitive, Agentic and Human–AI Collaboration layers combining real-time data, predictive analytics, autonomous execution and human supervision.	Predictive analytics; Agentic AI; real-time data; Human–AI collaboration. Strong architectural match for predictive-to-action workflows.	https://ieeexplore.ieee.org/document/11395643/
16	Dual-View Deep Learning Approach for Predictive Business Process Monitoring	Binbin Chen; Shuangyao Zhao; Qiang Zhang; Chunhua Tang; Leilei Lin	IEEE Transactions on Services Computing	18 April 2025	Predicts the next activity in an ongoing business process using both event-order information and event attributes.	LSTM self-attention + 12-layer ResNet; evaluated on 12 real-life event logs. Directly supports predictive business workflow intelligence.	https://ieeexplore.ieee.org/document/10970106/
17	A Machine Learning-Based Approach for Multi-Objective, Multi-Product, and Multi-Period Supply Chain Optimization via Demand Forecasting	Farshid Abdi; Shaghayegh Abolmakarem; Amir Karbassi Yazdi; Virgil Popescu; Ramona Birau; Adrian T. Mitroi	IEEE Access	18 August 2025	Combines demand forecasting with multi-objective supply-chain optimization and dynamically adjusts decisions using predicted	Wavelet-MLP; time-series decomposition; weighted-sum optimization; open-source CBC solver. Uses a real four-year demand dataset.	https://ieeexplore.ieee.org/document/11129066/

					demand.		
18	Towards Intelligent Enterprise Assistants: Leveraging RAG to Connect LLMs with Structured Business Data	Dat Dang Thanh; Dung Ha Nguyen; Binh Nguyen Thanh	2025 RIVF International Conference on Computing and Communication Technologies	18–20 December 2025	Builds a RAG pipeline that translates questions into executable SQL, retrieves current structured business data and integrates it with LLM responses.	RAG; text-to-SQL; relational databases; structured enterprise data. Reports up to 87.5% execution accuracy and 91.2% component matching.	https://ieeexplore.ieee.org/document/11365184/
19	Engineering Scalable AI Pipelines: A Cloud-Native Approach for Intelligent Transactional Systems	Venkata Reddy Pasam; Pooja Devaraju; Vijayalaxmi Methuku; Kalpan Dharamshi; et al.	2025 International Conference on Computing Technologies (ICOCT)	13–14 June 2025	Presents a modular cloud-native AI pipeline for transactional systems covering data ingestion, preprocessing, training and inference.	Microservices; container orchestration; serverless computing; Kubernetes. Supports scalable production AI workflow architecture.	https://ieeexplore.ieee.org/document/11118443/
20	Data-Driven IT/OT Cyber Risk Evaluation and Mitigation Prioritization Using Machine Learning	Authors as listed on IEEE Xplore	2025 IEEE Conference on Artificial Intelligence (CAI)	5–7 May 2025	Uses ML to assess risk, identify high-risk techniques and prioritize mitigation strategies. It connects prediction with recommended operational responses.	Feature engineering; MITRE ATT&CK; CVSS; statistical models; RNN. Useful as an example of prediction → prioritization → action.	https://ieeexplore.ieee.org/document/11050455/

3. Synthesis of Literature Findings

Predictive business workflows: Recent predictive-process research demonstrates that historical event traces and process attributes can be used to predict future process activities and outcomes. This provides a foundation for introducing a measurable ML prediction layer before automated decisions.

Agentic and multi-agent workflows: Recent IEEE work shows that complex tasks can be decomposed into specialized agents responsible for interpretation, analysis, verification, recommendation and execution. The literature supports modular agent orchestration rather than relying on a single unrestricted LLM call.

Explainability and trustworthy reasoning: Recent work combining SHAP/XAI with LLMs demonstrates a route from model-level feature importance to natural-language explanations. This supports using structured prediction evidence as input to business-oriented LLM reasoning.

Human-in-the-loop decision making: Human-in-multi-agent research shows that human feedback can be integrated into agentic workflows to refine decisions. This supports risk-based escalation and approval rather than requiring every action to be fully autonomous.

Enterprise data and action execution: Recent enterprise-assistant and API-agent research demonstrates the importance of connecting LLMs to structured business data and executable APIs. This supports a workflow architecture in which reasoning is connected to real business actions.

Auditability and provenance: PROV-AGENT demonstrates the importance of tracking agent interactions, decisions and downstream outcomes. This supports maintaining evidence and provenance for automated business decisions.

4. Identified Research Direction

The literature indicates that predictive ML, explainable AI, agentic reasoning, human oversight, enterprise data access and provenance are each active areas of research. The proposed project direction is to integrate these capabilities into one reusable business workflow rather than treating prediction, reasoning and execution as isolated components.

The research focus is therefore an outcome-oriented workflow in which a measurable prediction is used to guide an agentic business decision, with explanation, approval controls, execution and outcome measurement included in the same pipeline.

5. Proposed Representative Workflow

Proposed workflow for implementation and evaluation:

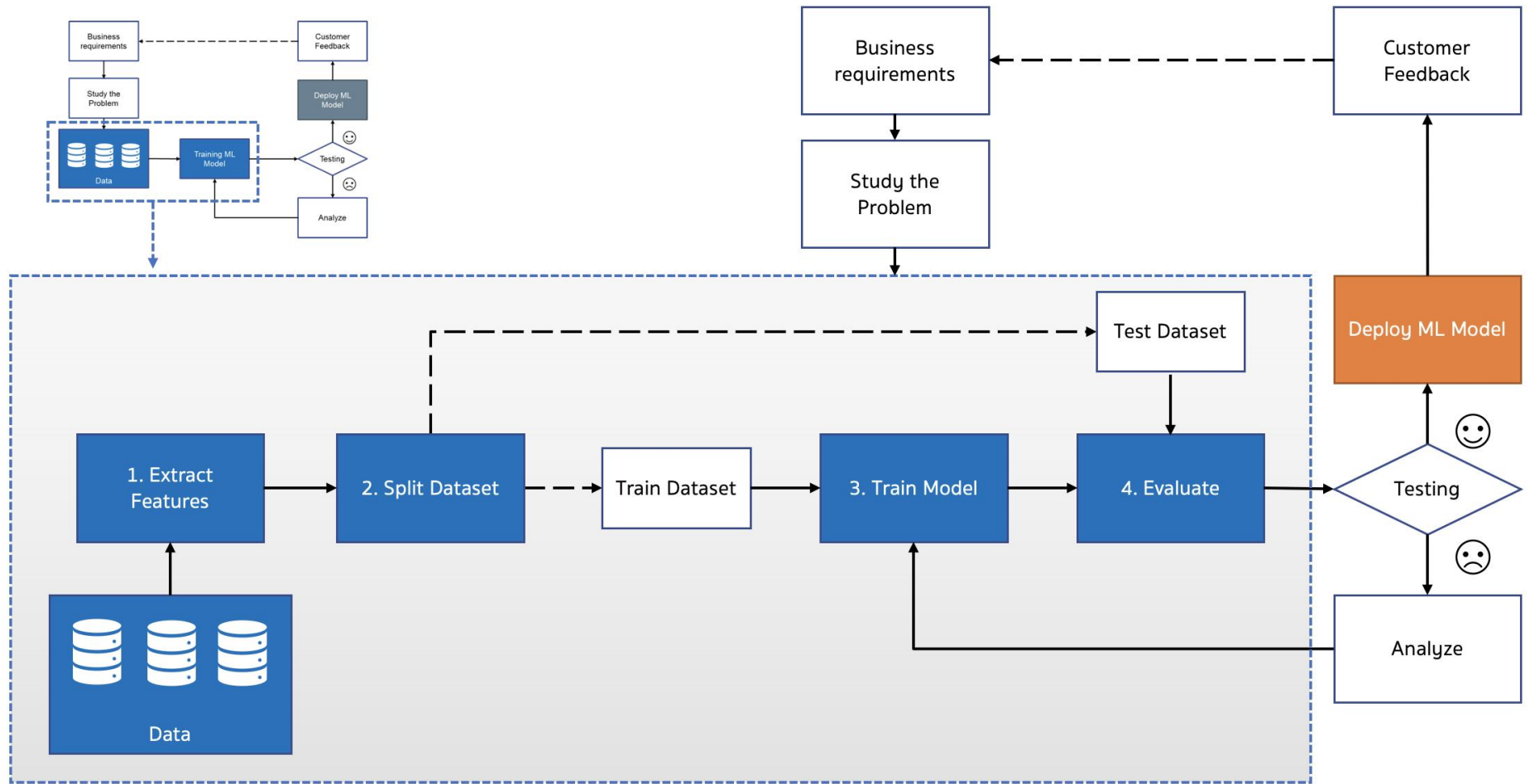
- AI-Powered Sales Lead Prioritization and Intelligent Follow-up Automation
- Identify and structure lead/customer interaction data
- Predict conversion probability or follow-up priority using ML
- Generate feature-level explanation for the prediction
- Use an LLM to interpret the structured prediction and prepare a follow-up recommendation
- Apply a risk/impact policy to decide between automatic execution and human approval
- Execute the approved action through an email/API integration

- Record response and conversion outcomes
- Evaluate business impact and use outcomes as feedback

6. Project Development and Progress Flow

Figure 1. Proposed Project Development and System Evolution Flow

Machine Learning Workflow



Existing Problem / Workflow Understanding



How Human-in-the-Loop AI Agents Work

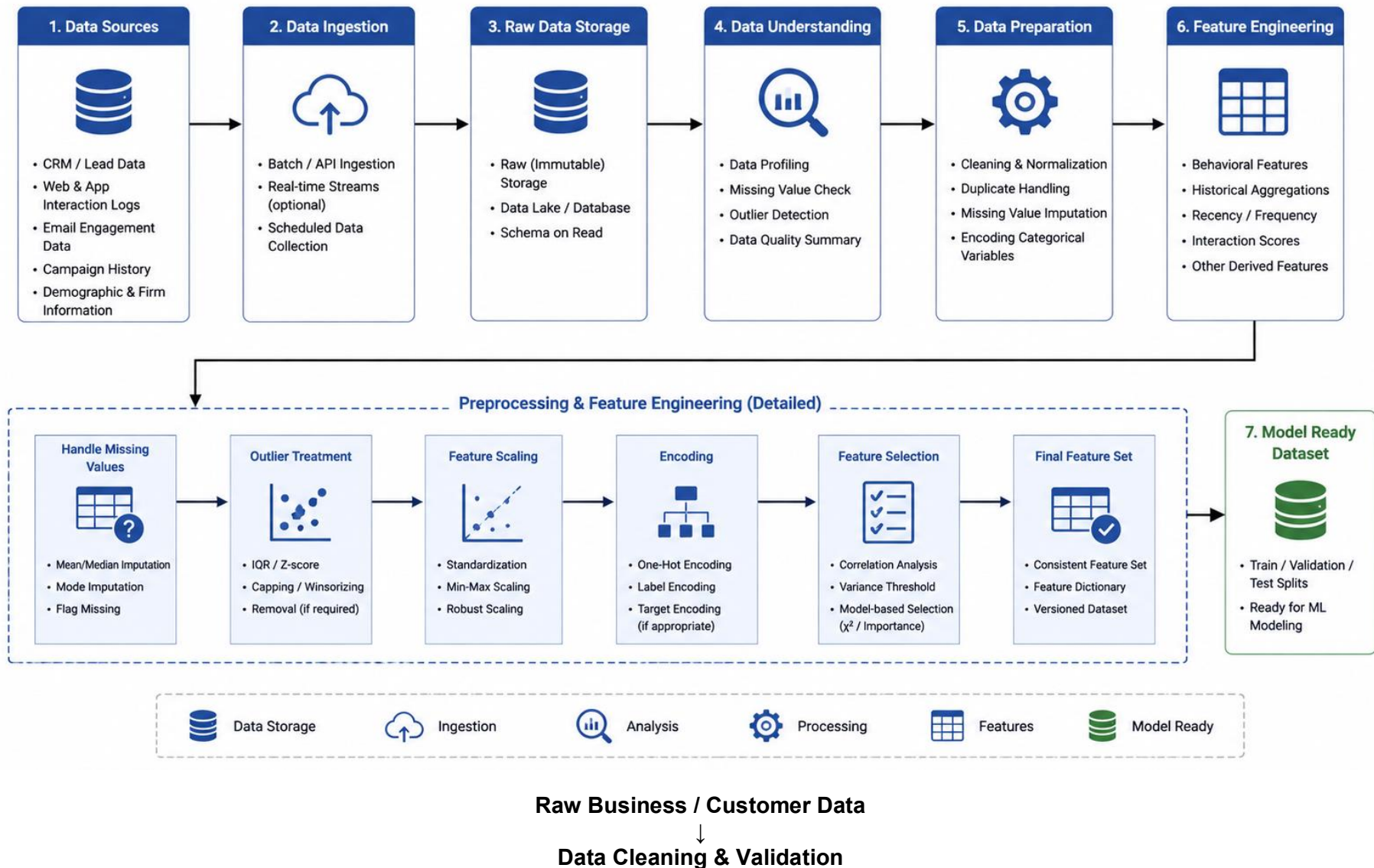


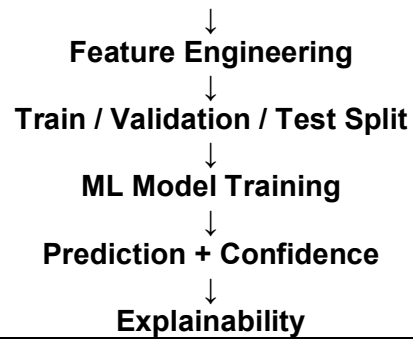
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7. Proposed Input Data and Dataset Strategy

Figure 2. Dataset and Input-Feature Pipeline

Figure 2. Dataset / Input Data Pipeline



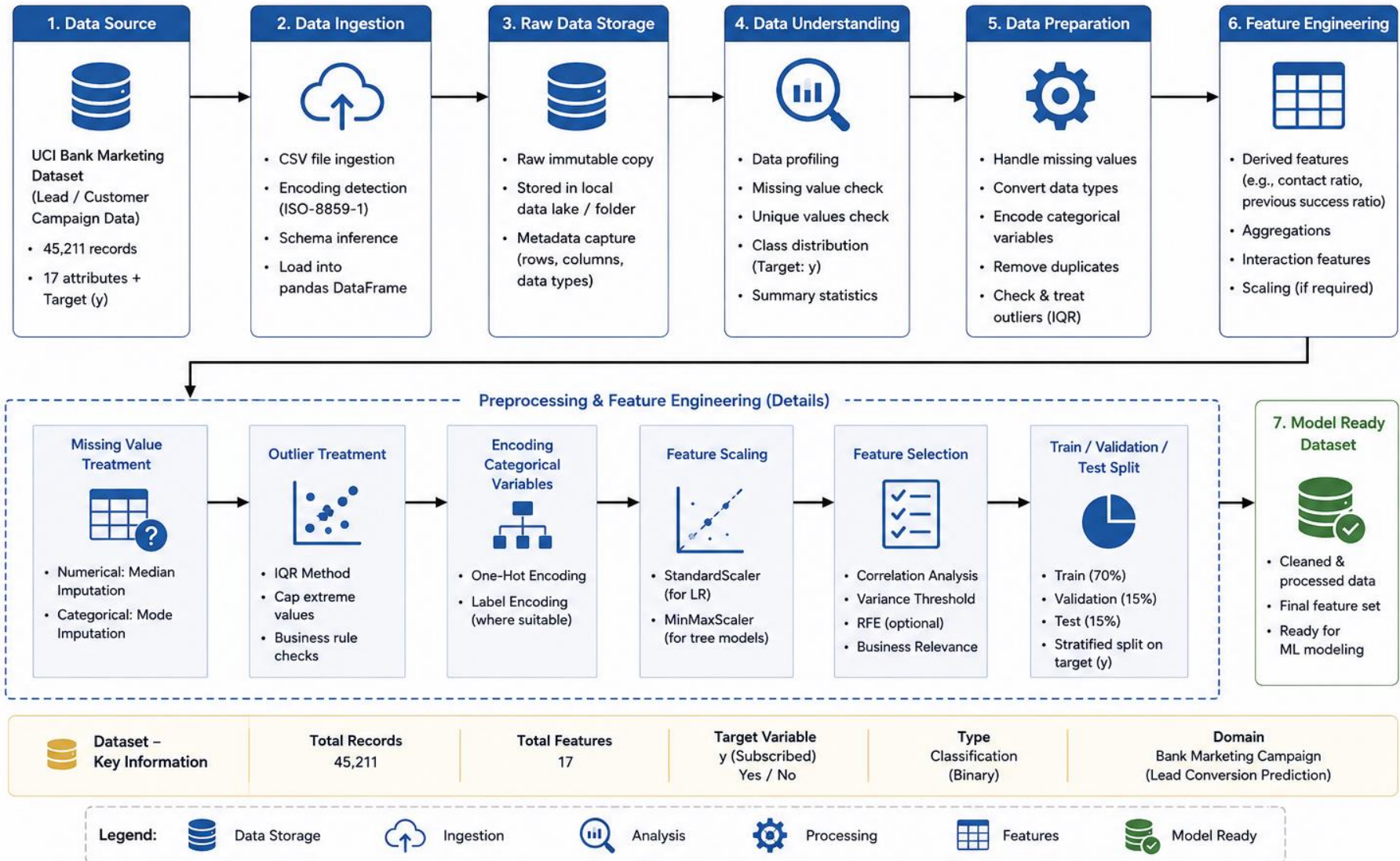


Proposed input categories for the representative sales workflow:

- Customer / lead profile information
- Campaign and contact history
- Previous interaction outcomes
- Recency and frequency of interaction
- Contact/channel information
- Historical conversion outcome (target variable)

The final feature list and dataset will be finalized after dataset validation. Only fields actually present in the selected dataset will be used for model training and evaluation.

Figure 2. Dataset / Input Data Pipeline
(Selected Data: Bank Marketing Dataset – Lead Prioritization Use Case)



Dataset / Input Data Description

The selected dataset for the proposed lead-prioritization workflow is the **Bank Marketing Dataset**, obtained from the UCI Machine Learning Repository. The dataset contains **45,211 customer records and 16 input attributes**, with the binary target variable y , indicating whether a customer subscribed to a term deposit. The available attributes cover customer demographics, financial characteristics, contact information, and previous campaign interaction history. These variables provide the basis for assessing the likelihood of customer conversion and prioritizing leads for subsequent follow-up actions.

For the proposed workflow, the dataset will be treated as a **lead-assessment and prioritization problem**. The preprocessing pipeline includes data validation, categorical-variable encoding, feature engineering, feature selection, and train/validation/test splitting before ML model development. The duration attribute will be excluded from the primary predictive model because it represents the duration of the current contact and is only known after the contact has taken place. Excluding it makes the prediction more representative of a real lead-prioritization scenario, where the objective is to estimate conversion likelihood **before or at the point of deciding whom to contact**. The UCI dataset documentation explicitly identifies this limitation for realistic predictive use.

The resulting model will generate a **conversion probability / lead-priority score** for each lead. These predictions can subsequently be combined with explainability methods and LLM-based reasoning to support personalized follow-up recommendations within the SMBFlow workflow.

Dataset Sources

Primary Dataset — UCI Machine Learning Repository

UCI Bank Marketing Dataset

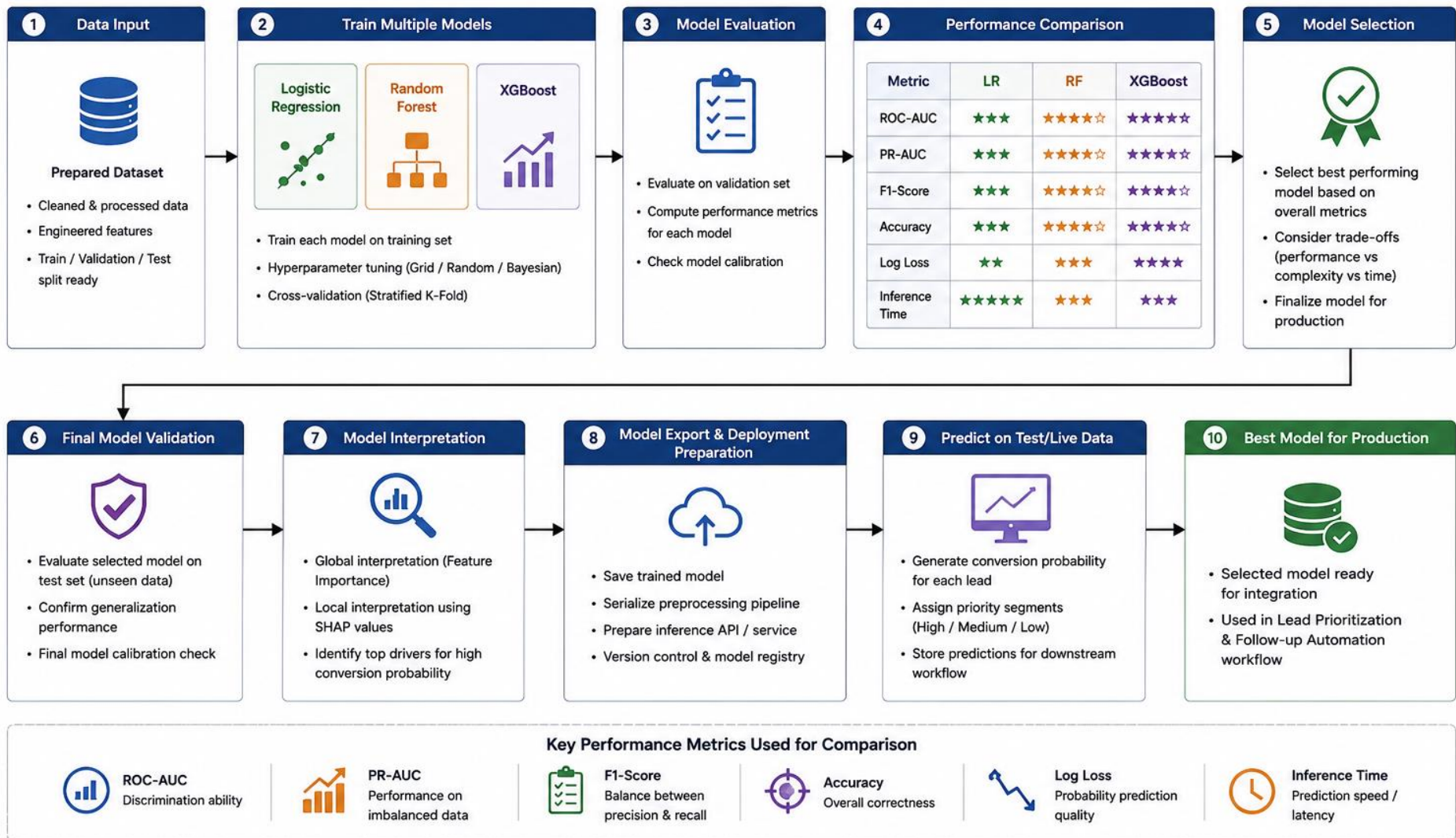
Alternative Dataset Mirror — Kaggle

Kaggle Bank Marketing Dataset

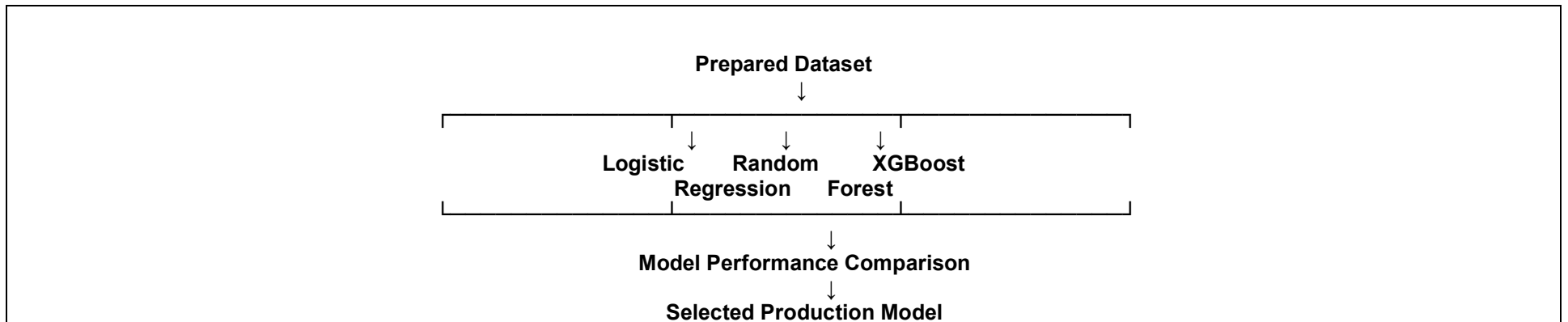
8. Multi-Model ML Development

Figure 3. Proposed Multi-Model Comparison Pipeline

Figure 3. Multi-Model Development, Comparison and Selection Pipeline
(Lead Prioritization – Bank Marketing Dataset)



Note: Final model selection is based on a combination of predictive performance, interpretability, calibration and inference efficiency.



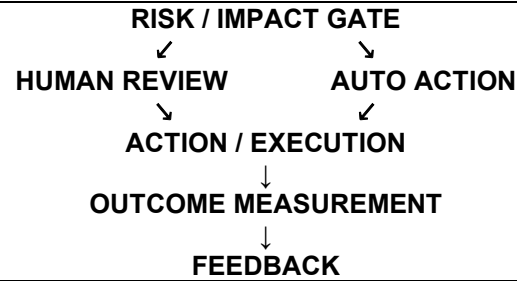
Planned evaluation measures:

- ROC-AUC
- PR-AUC
- F1-score
- Calibration / Brier score
- Inference time
- Top-K prioritization performance

9. Evolution of the Proposed Architecture

Figure 4. Proposed Architecture Pipeline





The proposed architecture separates prediction from language reasoning. The ML layer produces a measurable prediction, the explainability layer provides evidence for that prediction, and the LLM uses the structured evidence and business context to generate a recommendation. A policy layer then determines whether the action requires human approval or can be executed automatically.

10. Proposed Experimental Comparison

The evaluation will compare a conventional LLM-driven workflow against the proposed predictive-agentic workflow.

Evaluation Dimension	Baseline: LLM-Only Workflow	Proposed: ML + XAI + LLM Workflow
Prediction quality	Not applicable / rule or LLM output	Measured ML prediction
Explanation quality	LLM-generated explanation	ML evidence + XAI + LLM explanation
Token usage	Measured	Measured
Latency	Measured	Measured
Action accuracy	Measured	Measured
Human approval rate	Measured	Measured
Business outcome	Measured	Measured

11. Expected Measurable Outcomes

- Improved prioritization of high-probability opportunities
- Reduced time spent manually reviewing and prioritizing leads
- Reduced follow-up delay
- Improved follow-up consistency
- Conversion / response rate measurement
- Human approval rate for higher-risk actions
- LLM token and latency efficiency
- Traceable evidence for automated decisions

12. Research and Implementation Milestones

Stage	Main Activity	Current Status
1	Literature review and research-gap analysis	Completed / under refinement
2	Representative workflow selection	Under evaluation
3	Dataset identification and validation	To be finalized
4	Feature engineering and preprocessing	Planned
5	Multi-model ML baseline and comparison	Planned
6	Explainability layer	Planned
7	LLM reasoning and recommendation layer	Planned
8	Human-in-the-loop and action policy	Planned
9	Backend/API and workflow integration	Planned
10	Experimental and business evaluation	Planned

13. References / IEEE Xplore Sources

1. **Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey** — IEEE Access, 22 January 2025. <https://ieeexplore.ieee.org/document/10849561/>
2. **Agentic AI: A Survey of Autonomous Agents, Architectures, and Emerging Applications** — 2025 IEEE ICTBIG, 12–13 December 2025. <https://ieeexplore.ieee.org/document/11323264/>
3. **Alvaluate: A Multi-Agent Framework for Automated Answer Script Evaluation Using Large Language Models and Semantic Vector Indexing** — IEEE Access, 2025. <https://ieeexplore.ieee.org/document/11155081/>
4. **MAD-CTI: Cyber Threat Intelligence Analysis of the Dark Web Using a Multi-Agent Framework** — IEEE Access, 3 March 2025. <https://ieeexplore.ieee.org/document/10908603/>
5. **Towards Detecting LLMs Hallucination via Markov Chain-based Multi-agent Debate Framework** — ICASSP 2025, 6–11 April 2025. <https://ieeexplore.ieee.org/document/10889448/>
6. **AnaFlow: Agentic LLM-based Workflow for Reasoning-Driven Explainable and Sample-Efficient Analog Circuit Sizing** — 2025 IEEE/ACM ICCAD, 26–30 October 2025. <https://ieeexplore.ieee.org/document/11240818/>
7. **Generative Explainability for Next-Generation Networks: LLM-Augmented XAI with Mutual Feature Interactions** — 2025 IEEE WiMob, 20–22 October 2025. <https://ieeexplore.ieee.org/document/11257542/>
8. **A Human-in-Multi-Agent-Loop Intent Refinement Method for Task Offloading in Mobile Edge Computing** — 2025 IEEE ECIS, 23–25 May 2025. <https://ieeexplore.ieee.org/document/11086847/>
9. **PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions in Agentic Workflows** — 2025 IEEE eScience, 15–18 September 2025. <https://ieeexplore.ieee.org/document/11181558/>

10. PDD-AGENT: Multimodal Large Language Model-Driven AI Agent for Enhanced Plant Disease Diagnosis — 2025 IEEE ICIP, 14–17 September 2025.
<https://ieeexplore.ieee.org/document/11084359/>

11. RUBRA: An Agentic AI System for Automatic Short Answer Grading Using LLMs and RAG — 2025 IEEE conference publication, 2025.
<https://ieeexplore.ieee.org/document/11189329/>

12. Intent-Based Kubernetes Configuration via LLMs: Current Trends and Open Challenges — 2025 IEEE LCN, 13–16 October 2025.
<https://ieeexplore.ieee.org/document/11146359/>

13. The Rise of Agentic AI in Finance: Opportunities, Risks, and Human-Centric Integration — IEEE IT Professional, 14 August 2025.
<https://ieeexplore.ieee.org/document/11125703/>

14. Intelligent Cybersecurity Defense Strategies Based on Large Language Model Agents — 2025 IEEE ASENS, 21–23 March 2025.
<https://ieeexplore.ieee.org/document/11011130/>

15. Supply Chain with Sixth Sense Agentic AI: A Proactive Intelligence Framework for Autonomous Planning — 2026 IEEE ICAIC, 18–20 February 2026.
<https://ieeexplore.ieee.org/document/11395643/>

16. Dual-View Deep Learning Approach for Predictive Business Process Monitoring — IEEE Transactions on Services Computing, 18 April 2025.
<https://ieeexplore.ieee.org/document/10970106/>

17. A Machine Learning-Based Approach for Multi-Objective, Multi-Product, and Multi-Period Supply Chain Optimization via Demand Forecasting — IEEE Access, 18 August 2025. <https://ieeexplore.ieee.org/document/11129066/>

18. Towards Intelligent Enterprise Assistants: Leveraging RAG to Connect LLMs with Structured Business Data — 2025 RIVF International Conference on Computing and Communication Technologies, 18–20 December 2025. <https://ieeexplore.ieee.org/document/11365184/>

19. Engineering Scalable AI Pipelines: A Cloud-Native Approach for Intelligent Transactional Systems — 2025 International Conference on Computing Technologies (ICOCT), 13–14 June 2025. <https://ieeexplore.ieee.org/document/11118443/>

20. Data-Driven IT/OT Cyber Risk Evaluation and Mitigation Prioritization Using Machine Learning — 2025 IEEE Conference on Artificial Intelligence (CAI), 5–7 May 2025.
<https://ieeexplore.ieee.org/document/11050455/>

1. Risk-Adaptive Autonomy ★

Instead of fixed human approval rules, SMBFlow can **dynamically decide the level of autonomy** based on:

ML confidence + predicted business impact + action risk + historical action success

High confidence + Low risk

→ Auto Execute

Medium confidence / Medium risk

→ Human Approval

Low confidence / High risk

→ Human Decision



Novelty: The system doesn't just predict or automate; it **decides how much autonomy it should have for each business action.**

2. Prediction → Action Utility Scoring ★

Don't select an action only because the lead has high conversion probability.

Calculate something like:

Expected Action Utility = Probability of Success × Business Value – Action Cost/Risk

So:

Lead A: 90% conversion × ₹500 value = high utility

Lead B: 95% conversion × ₹20 value = lower utility

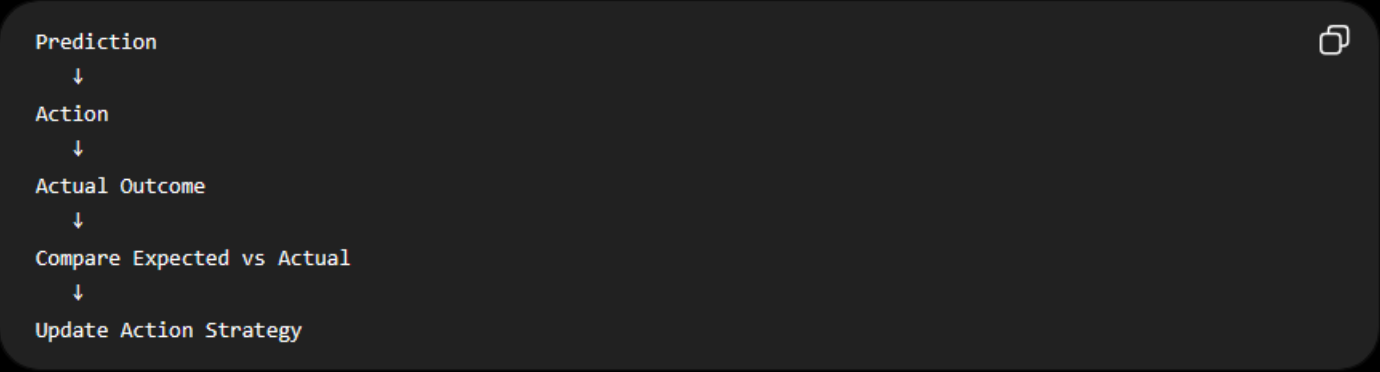


The system prioritizes the **action with maximum expected business value**, not simply maximum ML probability.

This is much stronger than basic lead prediction.

3. Outcome-Based Agent Feedback ★★ ★

After every automated action:



```
graph TD; A[Prediction] --> B[Action]; B --> C[Actual Outcome]; C --> D[Compare Expected vs Actual]; D --> E[Update Action Strategy];
```

Prediction
↓
Action
↓
Actual Outcome
↓
Compare Expected vs Actual
↓
Update Action Strategy

For example:

Model predicted 80% conversion → email sent → customer did not respond.

The system records this and learns **which recommendations/actions actually work**, instead of only retraining the prediction model.

This can become your strongest research contribution.